US Arrests: PCA and Clustering

Grouping the 50 US states by crime and urbanisation using **PCA**, **K-means** and **hierarchical clustering**. HyperionDev Unsupervised Machine Learning capstone project.

Data: US Arrests (1973): Murder, Assault and Rape arrests per 100,000 residents and % urban population.
Tools: Python, pandas, Matplotlib, seaborn, scikit-learn, SciPy, Jupyter

What I did

- Explored the data and standardised it, so Assault's large values wouldn't dominate.
- Ran a correlation analysis: the three crimes are strongly linked (0.56 to 0.80), and urbanisation is mostly separate.
- Applied PCA and kept 2 components, which explain **86.8%** of the variance. PC1 measures violent crime and PC2 measures urbanisation.
- Used the elbow method and silhouette scores to choose 4 clusters.
- Grouped the states with K-means (silhouette 0.444) and hierarchical clustering (0.413). Both methods placed 45 of the 50 states the same way.

Results

Cluster	Profile	Example states
0	High crime, rural	Mississippi, Alabama, Georgia
1	Low crime, rural	Maine, Vermont, North Dakota
2	Moderate crime, urban	Massachusetts, Ohio, New Jersey
3	High crime, urban	California, Florida, New York

Conclusion: States separate mainly by violent crime level, with urbanisation as a second factor. High crime wasn't only an urban problem: several rural southern states had high crime too.

Run it

```
pip install pandas matplotlib seaborn scikit-learn scipy jupyter
jupyter notebook Unsupervised_task.ipynb
```